

Failure Prevention

1. **Control:** Mandate Brake Override System (Smart Pedal) Requirements.
 - **SDLC Stage:** Requirements
 - **Type:** Preventive
 - **Fixes:** C1 (Requirements). Ensures that if both brake and throttle are depressed, the system forcefully cuts throttle at the hardware/OS level.
2. **Control:** Strict Adherence to MISRA C Coding Standards.
 - **SDLC Stage:** Implementation
 - **Type:** Preventive
 - **Fixes:** C4 (Implementation). Bans recursion, limits global variables, and reduces cyclomatic complexity to prevent memory corruption.
3. **Control:** Hardware ECC (Error-Correcting Code) RAM for Safety-Critical ECUs.
 - **SDLC Stage:** Design/Architecture
 - **Type:** Preventive
 - **Fixes:** C3 (Design). Prevents single-event upsets (cosmic rays/electromagnetic interference) from flipping bits in throttle control variables.
4. **Control:** RTOS Fault Injection Testing.
 - **SDLC Stage:** Testing
 - **Type:** Detective
 - **Fixes:** C5 (Testing). Simulates CPU overloads and stack overflows to ensure the fail-safe watchdog timer actually resets the system during a task death.
5. **Control:** Independent Safety Whistleblower Channel.
 - **SDLC Stage:** Operations / Management

- **Type:** Detective
- **Fixes:** C9/C10 (Management/Ethics). Allows engineers to bypass middle management and report unsafe architectural decisions directly to a safety board or regulators.

Sources Consulted - NASA Engineering & Safety Center / NHTSA technical reports (2011)

Github Link - [Lulu-Ping/lab01-failure-forensics1](https://github.com/Lulu-Ping/lab01-failure-forensics1)